

Cyber Security Trends

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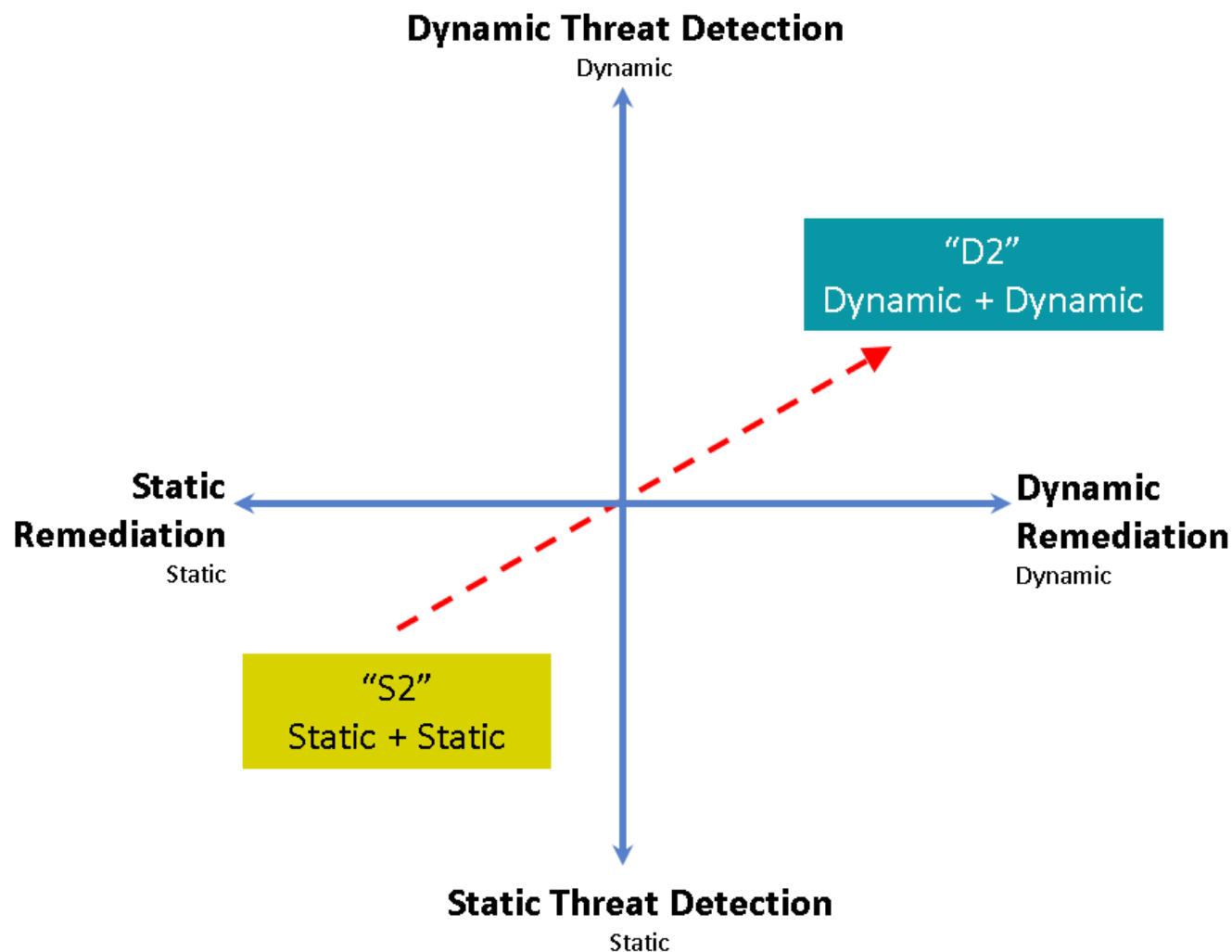
Current State

Introduction

- The Challenge:
 - Detection windows are shrinking
 - New attacks & vectors not previously identified
- Ransomware
 - # 1 Cyber-threat today
 - Ransomware Attacks are still penetrating defenses
- Attacks have become primarily Dynamic, rather than Static

Current State

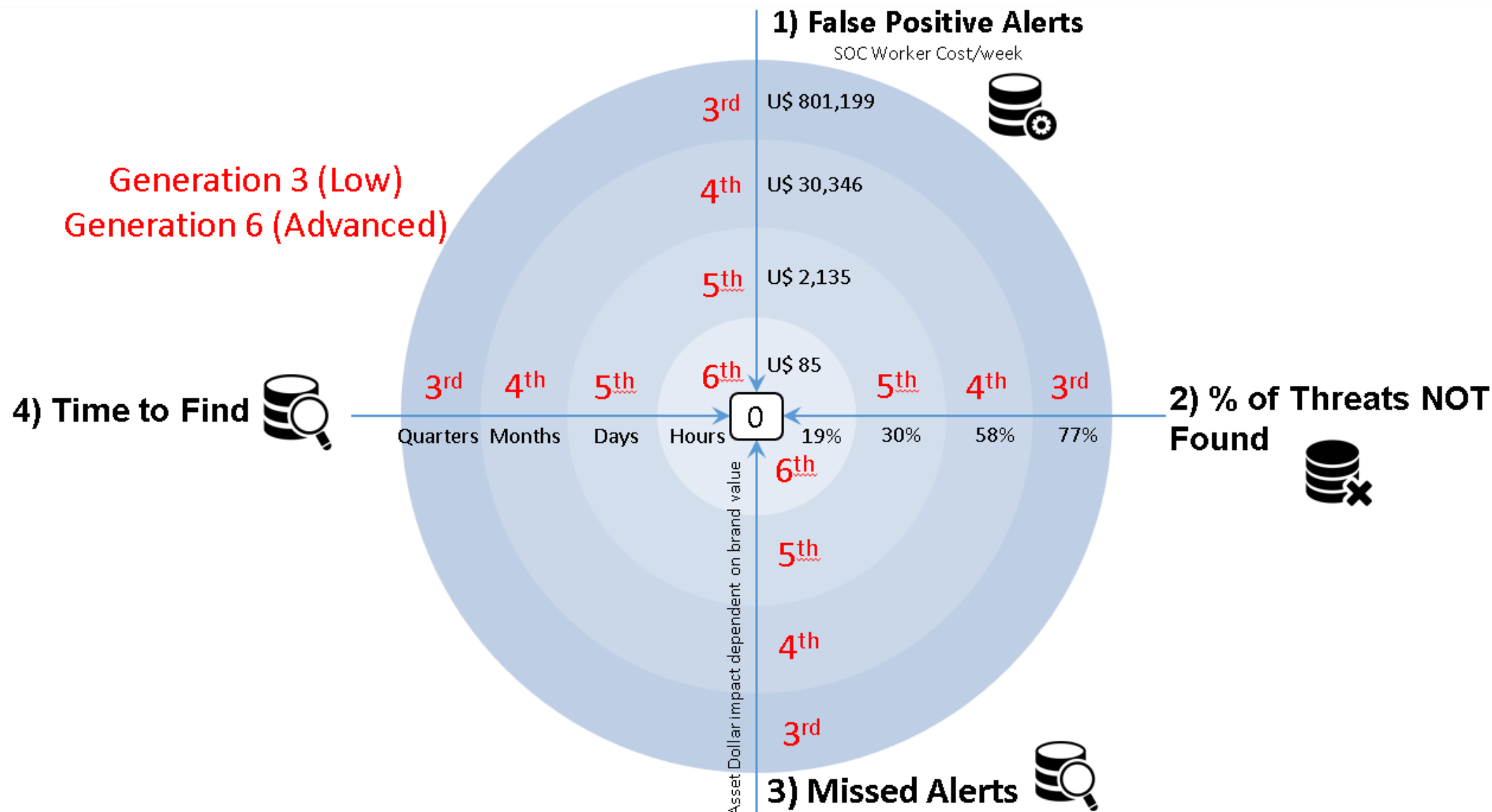
Introduction



- **40% of Threats Today Are Static**
 - They are Known Threats
 - Recognizable/Known
 - Do not regularly change, evolve, or transform
- **60% of Threats Today Are Dynamic**
 - They are Unknown Threats
 - Very hard to Detect
 - Change Constantly
 - Cannot be identified using signatures, IOC's, Rules
 - Simulate "normal" behavior

Current State

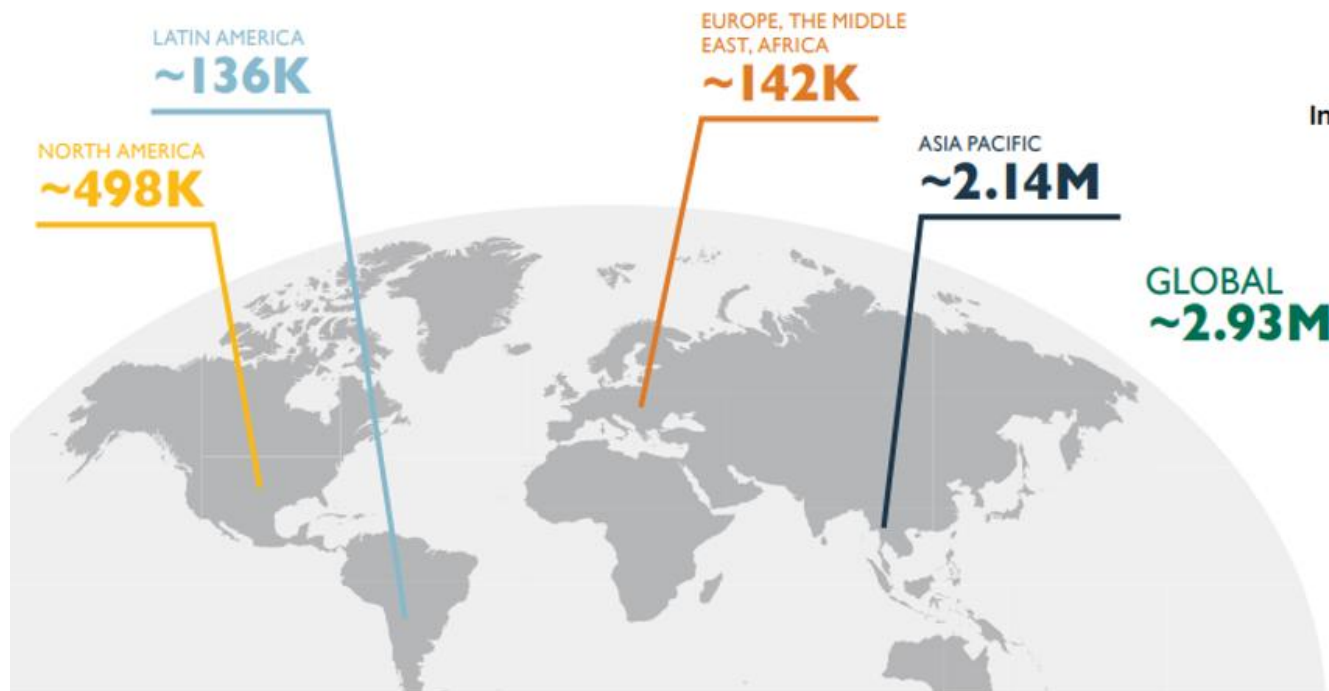
The Race to Zero



Current State

Talent Shortage and High Cost

Gap in Cybersecurity Professionals by Region



Information Security Analysts

Median annual wages, May 2018

Information security analysts

\$98,350

Trying to build your own SOC with all the security tools and staff can well cost over \$1M and ongoing cost upward of \$500k per year, if not more



As of 2024 there were close to 700K in Cyber job openings that were not being filled

Source: <https://www.peoplemattersglobal.com/news/talent-conversation/asia-pacifics-cybersecurity-workforce-gap-hits-214m-19654>

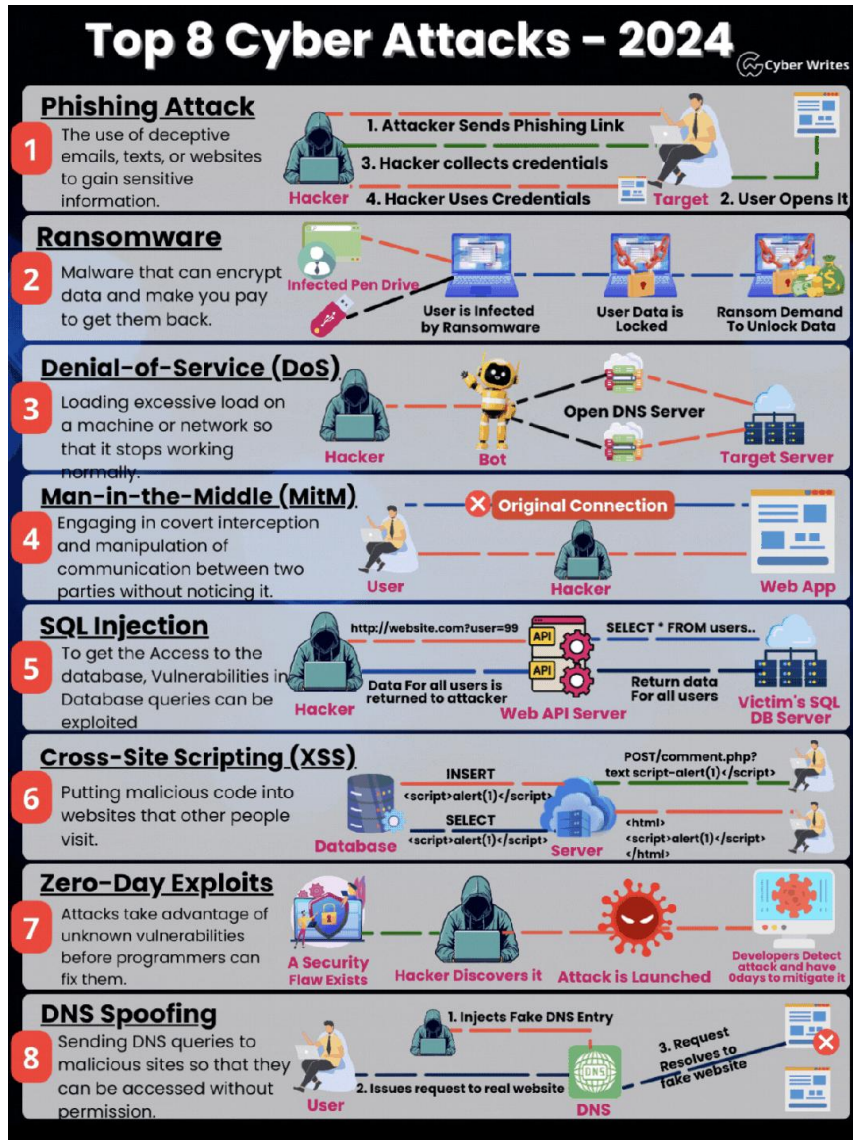
Current State

Steps to Design Secure Systems

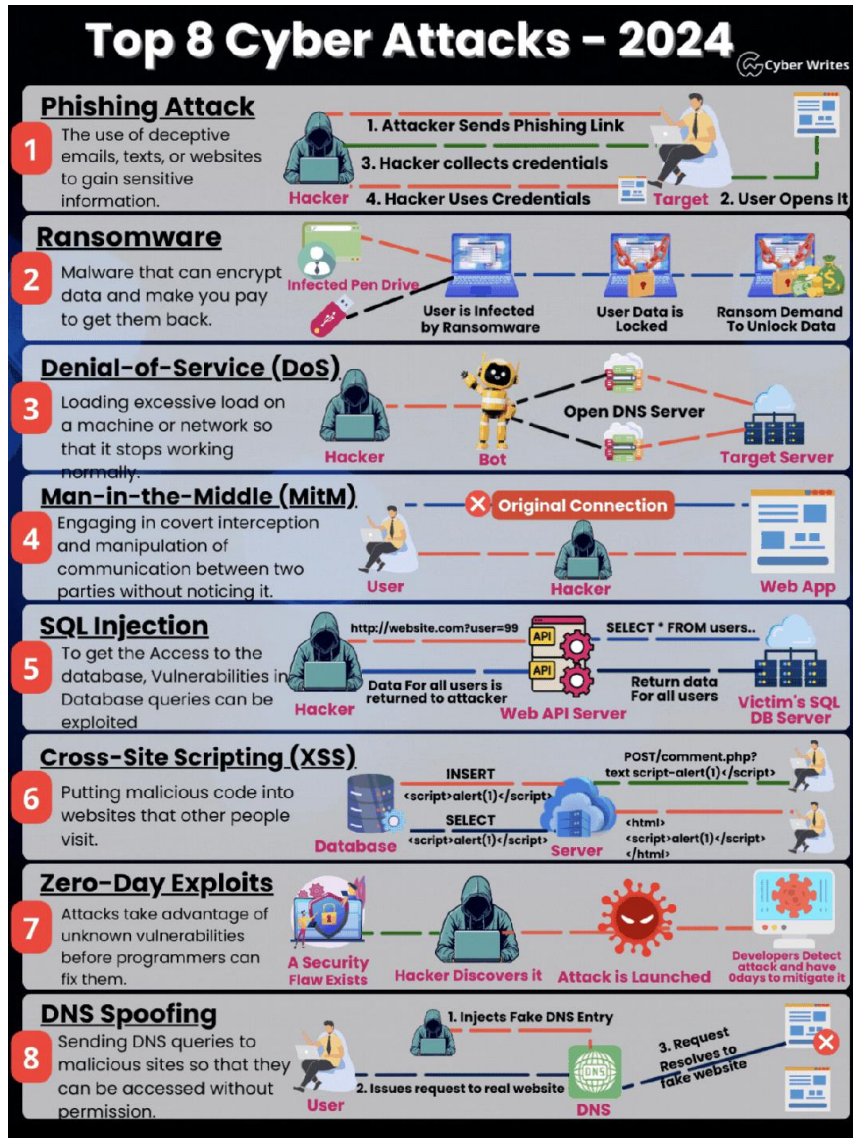


Trends

Top Cyber Attacks



Top Cyber Attacks



PORTS MOSTLY USED BY HACKERS

1. Port 21 - File Transfer Protocol (FTP)
2. Port 22 - Secure Shell (SSH)
3. Port 23 - Telnet
4. Port 25 - Simple Mail Transfer Protocol (SMTP)
5. Port 53 - Domain Name System (DNS)
6. Port 69 - TFTP
7. Port 80, 443, 8080, 8443 - HTTP/HTTPS
8. Port 135 - Windows RPC @Security Trybe
9. Port 137-139 - Windows NetBIOS over TCP/IP

Common Types of Attacks

1. [Malware](#)
2. [Denial-of-Service \(DoS\) Attacks](#)
3. [Phishing](#)
4. [Spoofing](#)
5. [Identity-Based Attacks](#)
6. [Code Injection Attacks](#)
7. [Supply Chain Attacks](#)
8. [Social Engineering Attacks](#)
9. [Insider Threats](#)
10. [DNS Tunneling](#)
11. [IoT-Based Attacks](#)
12. [AI-Powered Attacks](#)

Common Types of Attacks

Phishing

Type	Description
Spear Phishing	Spear-phishing is a type of phishing attack that targets specific individuals or organizations typically through malicious emails. The goal of spear phishing is to steal sensitive information such as login credentials or infect the targets' device with malware.
Whaling	A whaling attack is a type of social engineering attack specifically targeting senior or C-level executive employees with the purpose of stealing money or information, or gaining access to the person's computer in order to execute further cyberattacks.
SMiShing	Smishing is the act of sending fraudulent text messages designed to trick individuals into sharing sensitive data such as passwords, usernames and credit card numbers. A smishing attack may involve cybercriminals pretending to be your bank or a shipping service you use.
Vishing	Vishing, a voice phishing attack, is the fraudulent use of phone calls and voice messages pretending to be from a reputable organization to convince individuals to reveal private information such as bank details and passwords.

Common Types of Attacks

Spoofing

- Spoofing is a technique through which a cybercriminal **disguises themselves** as a known or **trusted source**.
- Goal: stealing information, extorting money or installing malware.

Type	Description
Domain Spoofing	Domain spoofing is a form of phishing where an attacker impersonates a known business or person with fake website or email domain to fool people into trusting them. Typically, the domain appears to be legitimate at first glance, but a closer look will reveal subtle differences.
Email Spoofing	Email spoofing is a type of cyberattack that targets businesses by using emails with forged sender addresses. Because the recipient trusts the alleged sender, they are more likely to open the email and interact with its contents, such as a malicious link or attachment.
ARP Spoofing	Address Resolution Protocol (ARP) spoofing or ARP poisoning is a form of spoofing attack that hackers use to intercept data. A hacker commits an ARP spoofing attack by tricking one device into sending messages to the hacker instead of the intended recipient. This way, the hacker gains access to your device's communications, including sensitive data.

Common Types of Attacks

Identity-based attacks

Type	Description
Kerberoasting	Kerberoasting is a post-exploitation attack technique that attempts to crack the password of a service account within the Active Directory (AD) environment. In a Kerberoasting attack, an adversary masquerades as an account user with a service principal name (SPN) and requests a ticket, which contains an encrypted password.
Man-in-the-Middle (MITM) Attack	A man-in-the-middle attack is a type of cyberattack in which an attacker eavesdrops on a conversation between two targets with the goal of collecting personal data, passwords or banking details, and/or to convince the victim to take an action such as changing login credentials, completing a transaction or initiating a transfer of funds.
Pass-the-Hash Attack	Pass the hash (PtH) is a type of attack in which an adversary steals a "hashed" user credential and uses it to create a new user session on the same network. It does not require the attacker to know or crack the password to gain access to the system. Rather, it uses a stored version of the password to initiate a new session.
Golden Ticket Attack	In a golden ticket attack, adversaries attempt to gain unlimited access to an organization's domain by accessing user data stored in Microsoft Active Directory. The attacker exploits vulnerabilities in the Kerberos identity authentication protocol, allowing them to bypass authentication methods.
Silver Ticket Attack	A silver ticket is a forged authentication ticket often created when an attacker steals an account password. A forged service ticket is encrypted and enables access to resources for the specific service targeted by the silver ticket attack.
Credential Harvesting	In credential harvesting, cybercriminals gather user credentials — such as user IDs, email addresses, passwords, and other login information — en masse to then access systems, gather sensitive data, or sell it in the dark web.
Credential Stuffing	Credential stuffing attacks work on the premise that people often use the same user ID and password across multiple accounts. Therefore, possessing the credentials for one account may be able to grant access to other, unrelated account.
Password Spraying	The basics of a password spraying attack involve a threat actor using a single common password against multiple accounts on the same application. This avoids the account lockouts that typically occur when an attacker uses a brute force attack on a single account by trying many passwords.
Brute Force Attacks	A brute force attack is uses a trial-and-error approach to systematically guess login info, credentials, and encryption keys. The attacker submits combinations of usernames and passwords until they finally guess correctly.
Downgrade Attacks	Downgrade attacks are a cyberattack where adversaries take advantage of a system's backward compatibility to force it into less secure modes of operation, such as forcing a user to go into a HTTP version of a website instead of HTTPS.

Common Types of Attacks

Social Engineering

- **Social engineering** is a technique where attackers use psychological tactics to manipulate people into taking a desired action.
- [Top 10 Social Engineering Attacks](#)

Attack	Description
Pretexting	In pretexting, attackers get access to information, a system, or a user by posing a false scenario that gains the victim's trust. This includes posing as an investment banker, an HR employee, or even an IT specialist.
Business email compromise (BEC)	In a BEC attack, attackers assume the identity of a trusted user to trick employees or company customers into making payments or sharing data, among other things.
Disinformation campaign	Disinformation campaigns are deliberate efforts to spread false information, especially for political or war-related reasons. Adversaries leverage social media networks that reach vast audiences to amplify false narratives through the prolific use of bots and fake accounts, creating a false sense of consensus.
Quid Pro Quo	Using the quid pro quo technique, attackers target users with an offer to pay in exchange for a product or service.
Honeytrap	Honeytrap attacks target individuals looking for love or friendship on dating apps/websites. Attackers create fake profiles and leverage the relationship built overtime to trick the victim into giving them money, information, or access to their network to install malware.
Tailgating/Piggybacking	Tailgating, also known as piggybacking, is a type of attack conducted in person by tagging along behind a company employee and asking them to hold the door open. Once the adversary is inside the facilities, they physically try to steal or destroy sensitive information.

Common Types of Attacks

AI-Powered Attacks

Attack	Description
Adversarial AI/ML	Adversarial artificial intelligence and machine learning seek to disrupt the operations of AI and ML systems by manipulating or misleading them. They can do this by introducing inaccuracies in training data.
Dark AI	Dark AI is specifically engineered to leverage the benefits of incorporating AI and ML technology to exploit vulnerabilities. Dark AI usually goes unnoticed until the damage is done.
Deepfake	Deepfakes are AI-generated forgeries that appear very real and have the potential to reshape public opinion, damage reputations, and even sway political landscapes. These can come in the form of fake images, videos, audio, or more.
AI-generated social engineering	Attackers create fake chatbots or virtual assistants capable of having human-like interactions and engaging in conversations with users to get them to provide sensitive information.

Common Types of Attacks

Malware

Type	Description
Ransomware	In a ransomware attack, an adversary encrypts a victim's data and offers to provide a decryption key in exchange for a payment. Ransomware attacks are usually launched through malicious links delivered via phishing emails, but unpatched vulnerabilities and policy misconfigurations are used as well.
Fileless Malware	Fileless malware is a type of malicious activity that uses native, legitimate tools built into a system to execute a cyber attack. Unlike traditional malware, fileless malware does not require an attacker to install any code on a target's system, making it hard to detect.
Spyware	Spyware is a type of unwanted, malicious software that infects a computer or other device and collects information about a user's web activity without their knowledge or consent.
Adware	Adware is a type of spyware that watches a user's online activity in order to determine which ads to show them. While adware is not inherently malicious, it has an impact on the performance of a user's device and degrades the user experience.
Trojan	A trojan is malware that appears to be legitimate software disguised as native operating system programs or harmless files like free downloads. Trojans are installed through social engineering techniques such as phishing or bait websites. The zeus trojan malware , a variant, has the goal accessing financial information and adding machines to a botnet.
Worms	A worm is a self-contained program that replicates itself and spreads its copies to other computers. A worm may infect its target through a software vulnerability or it may be delivered via phishing or smishing. Embedded worms can modify and delete files, inject more malicious software, or replicate in place until the targeted system runs out of resources.
Rootkits	Rootkit malware is a collection of software designed to give malicious actors control of a computer network or application. Once activated, the malicious program sets up a backdoor exploit and may deliver additional malware. Bootkits take this a step further by infecting the master boot prior to the operating system booting up, making them harder to detect.
Mobile Malware	Mobile malware is any type of malware designed to target mobile devices. Mobile malware is delivered through malicious downloads, operating system vulnerabilities, phishing, smishing, and the use of unsecured WiFi.
Exploits	An exploit is a piece of software or data that opportunistically uses a defect in an operating system or an app to provide access to unauthorized actors. The exploit may be used to install more malware or steal data.
Scareware	Scareware tricks users into believing their computer is infected with a virus. Typically, a user will see scareware as a pop-up warning them that their system is infected. This scare tactic aims to persuade people into installing fake antivirus software to remove the "virus." Once this fake antivirus software is downloaded, then malware may infect your computer.
Keylogger	Keyloggers are tools that record what a person types on a device. While there are legitimate and legal uses for keyloggers, many uses are malicious. In a keylogger attack, the keylogger software records every keystroke on the victim's device and sends it to the attacker.
Botnet	Botnet is a network of computers infected with malware that are controlled by a bot herder. The bot herder is the person who operates the botnet infrastructure and uses the compromised computers to launch attacks designed to crash a target's network, inject malware, harvest credentials, or execute CPU-intensive tasks.
MALSPAM	Malicious malware (MALSPAM) delivers malware as the malicious payload via emails containing malicious content, such as viruses or malware-infected attachments.
Wiper Attack	A wiper attack is designed to permanently delete or corrupt data on targeted systems. They are often observed in geopolitical conflicts and in a hacktivism context.

Ransomware

Ransomware Costs to Business

66% of organizations lost revenue following a ransomware attack.



32% of organizations lost C-Level talent as a direct result of ransomware attacks.

35% of businesses that paid a ransom demand shelled out between \$350,000 to \$1.4 million, while 7 percent paid ransoms exceeding \$1.4 million.



29% laid off employees due to financial pressures following a ransomware attack.

53% of organizations' brand and reputation were damaged in a successful attack.



26% of organizations were forced to close down operations entirely following a ransomware attack.

Ransomware

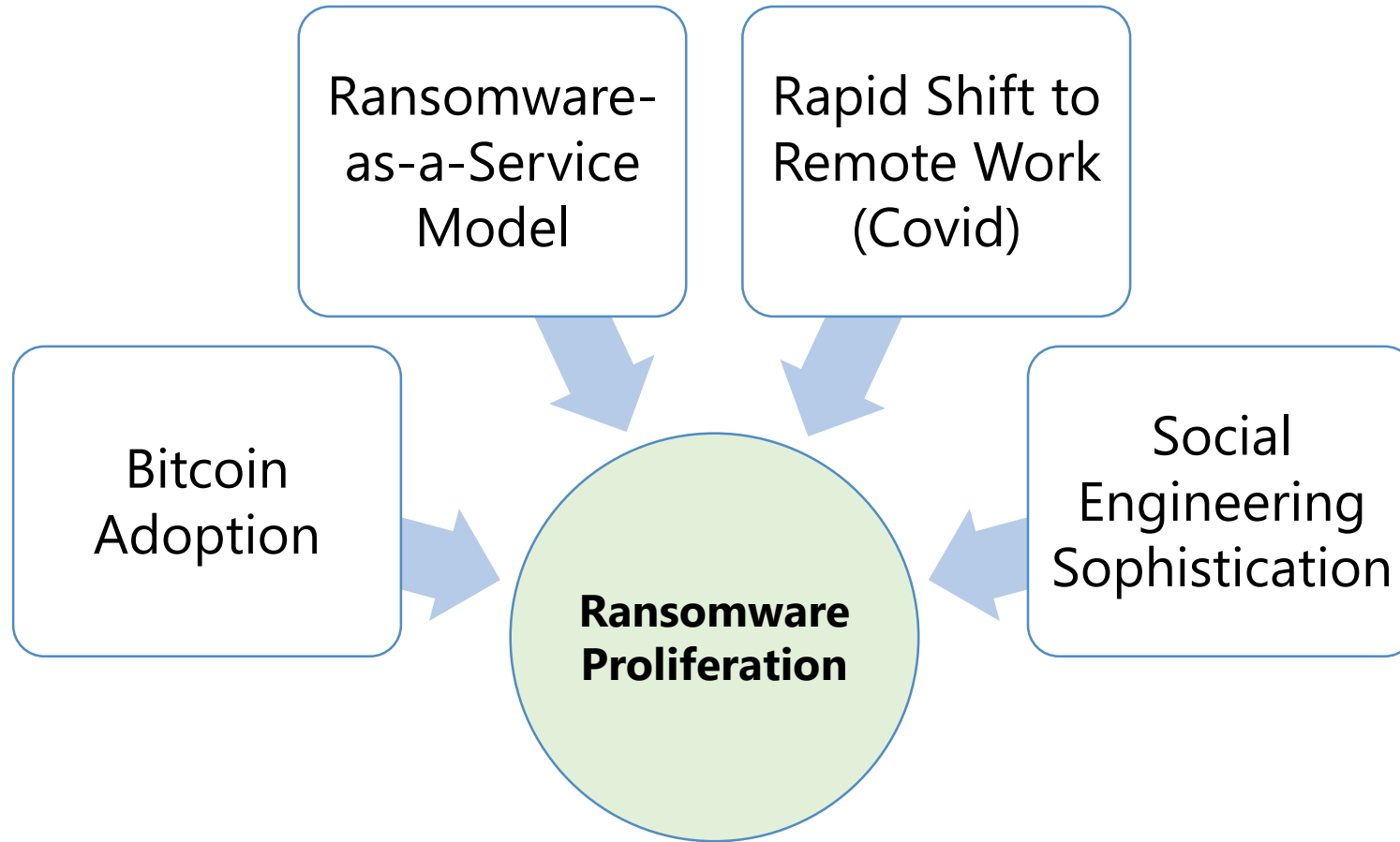
Why Now?

- Cyber-Criminal Focus is Shifting
- Previously
 - Credit Card & Identity Theft
 - Complacency Developed
- Ransom in exchange for computer access is not new
 - AIDS Trojan – Developed in 1989, Joseph Popp
 - Encrypting ransomware returned to prominence in late 2013 with the propagation of [CryptoLocker](#)
 - Bitcoin Used for first time
 - Procured \$27M from infected users
 - Became widely copied
- What is unique about now?



Ransomware

Multiple Factors Come Together



The combination of these technologies are responsible for the rapid scaling of a new cyber-criminal eco-system.

Ransomware

Criminal Eco-System is Expanding

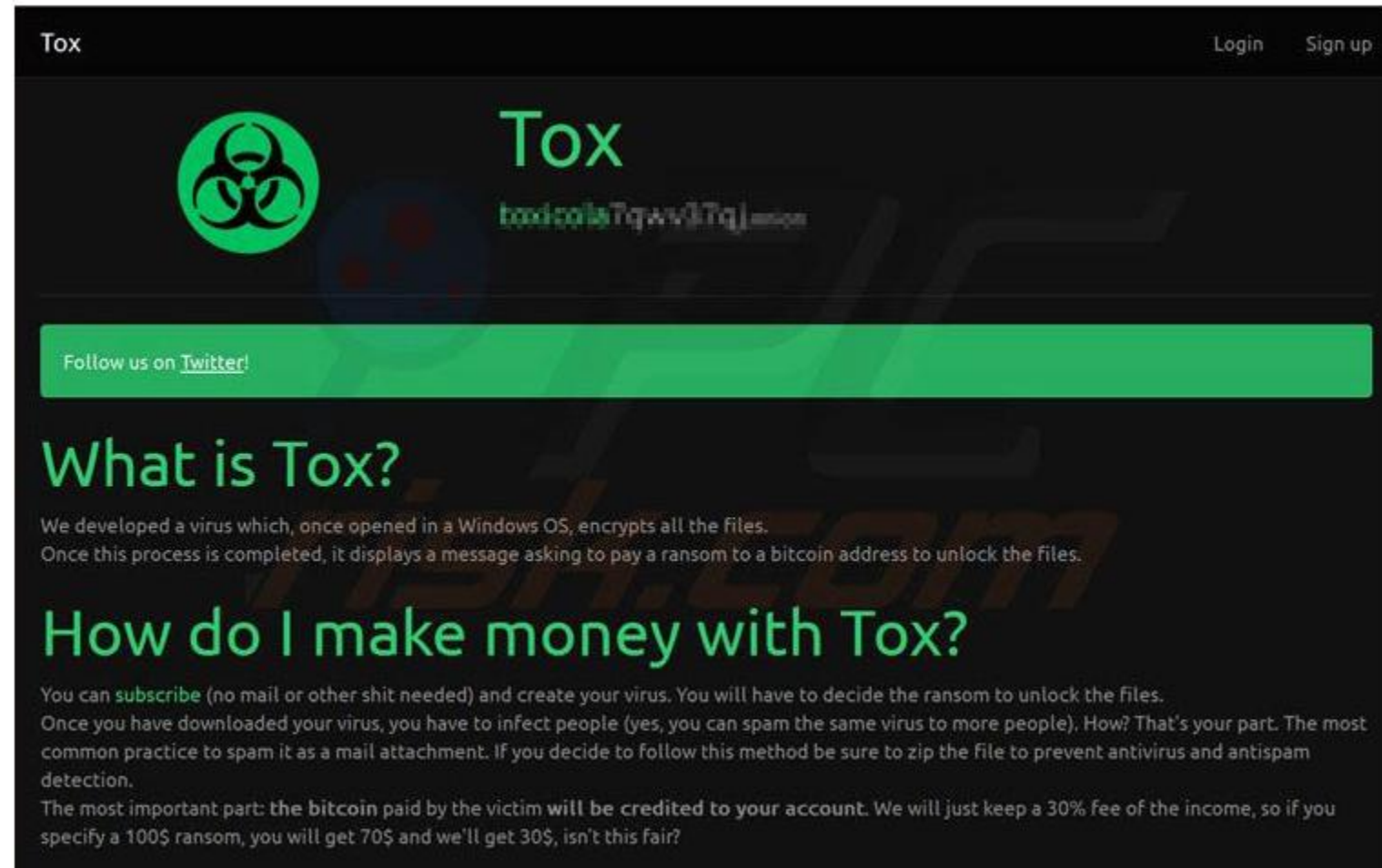
- The attackers **feel justified** in what they are doing
 - Wealth Disparity – (Robin hood syndrome)
 - “Ransomware is the Guillotine”
 - MountLocker's operator says: "We use network security holes the same way lawyers use legal holes."
- Ransomware is Scalable
 - If criminals, operating remotely, can make massive profits while incurring little risk to themselves, why would they ever stop?
- State Sponsored Attacks
 - It's not all for profit
 - Some governments are getting criminal gangs to do their hacking for them.



Ransomware

Ransomware as a Service (RaaS)

- Makes it easy for anyone to launch an attack.
- Fairly large criminal ecosystem.
- The attacker gets to set the ransom amount \$\$
- Disgruntled employee
 - Provides attack vector against the company
 - Receives the money from attack



Ransomware

How you get Infected?

- The top 4 exploited paths or vectors are:
 1. RDP/Remote Access (Stolen Credentials/Elevated Privileges)
 2. Phishing/Social Engineering
 3. Software Vulnerabilities/Lack of Patches
 4. Lateral Movement - SMB
- Any/All of these can be used in combination
 - Note: Are Coupled with other forms of malware that establish the beachhead
 - For example Trickbot

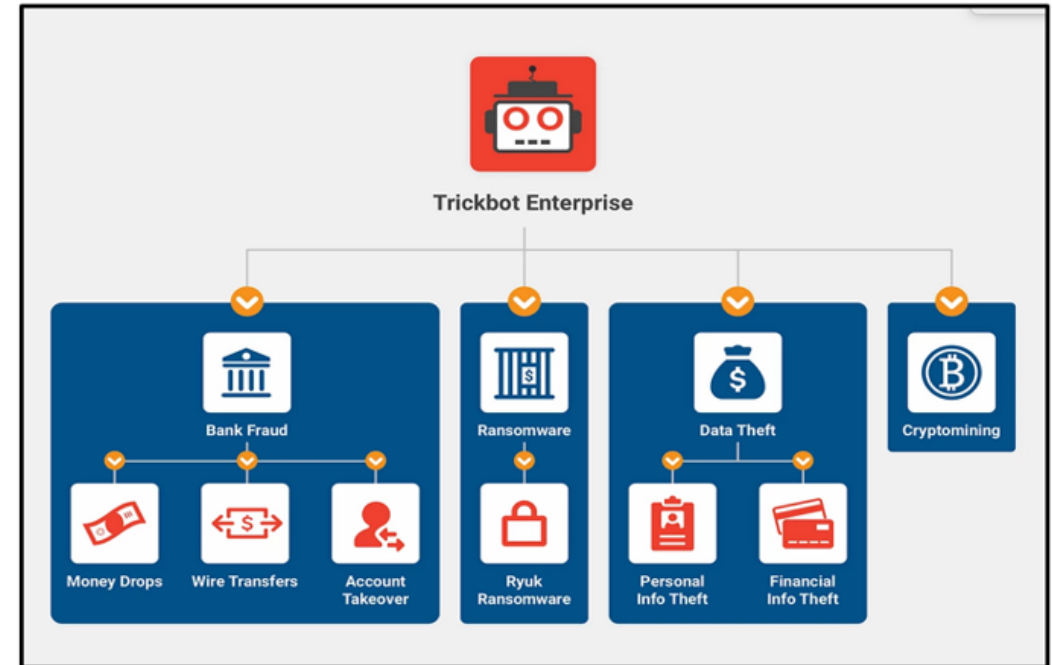
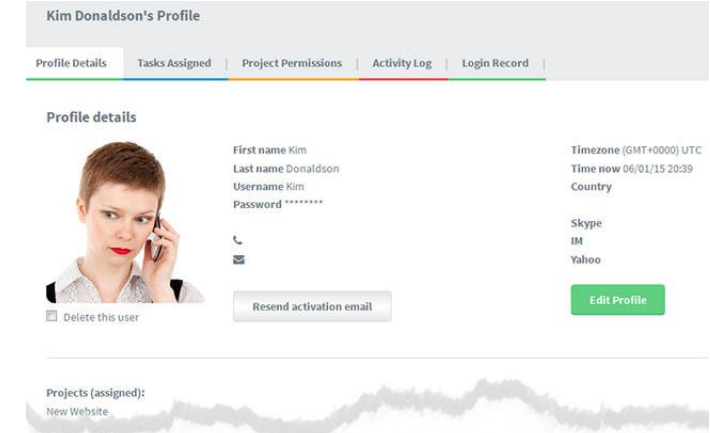


Figure 1: Overview of TrickBot's enterprise model. Source: Bleeping Computer

Ransomware

How to Detect?

- It conducts itself like a “normal user”
 - Reads, writes, opens files, creates files, lookups, downloads, etc.
- Speed of the Attack
 - Hackers used to sit in your system for months, sometimes year – No Longer
 - Attackers have already breached the network/endpoints/user access/etc.
 - It's the “end” phase of an attack - Once the encryption happens, you're too late
- Static Detection Methods are ineffective
 - Signatures, IOCs, Rules
 - New variations coming out constantly

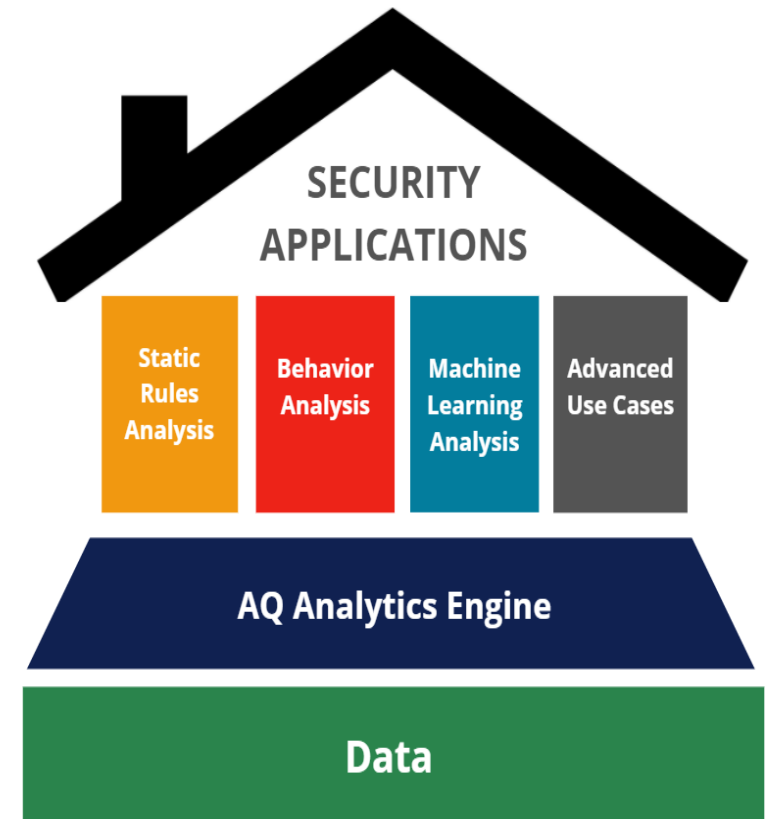


The Data Problem

The Data Problem

Introduction

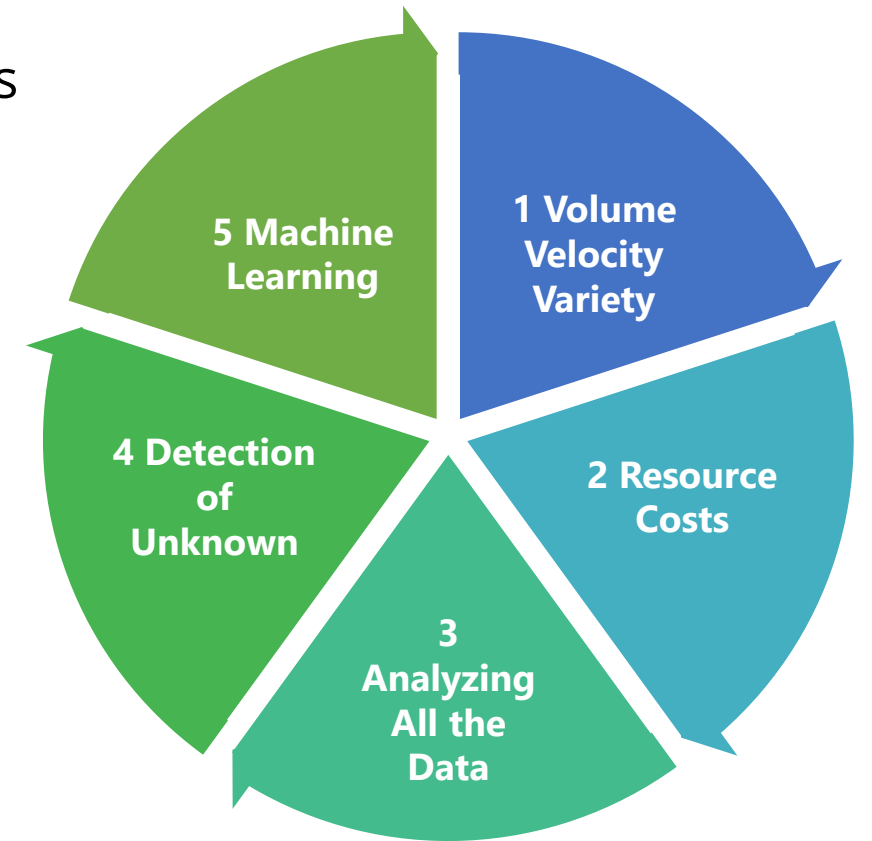
- Data access (speed & accuracy) is the inhibiting factor in threat analysis today.
- The Analytics Engine continuously analyzes all the data and then optimizes it for Machine Learning analysis.
- Analytics Engines core focus has been to find that threats that no one else can find in the data.



The Data Problem

Big Data Issues

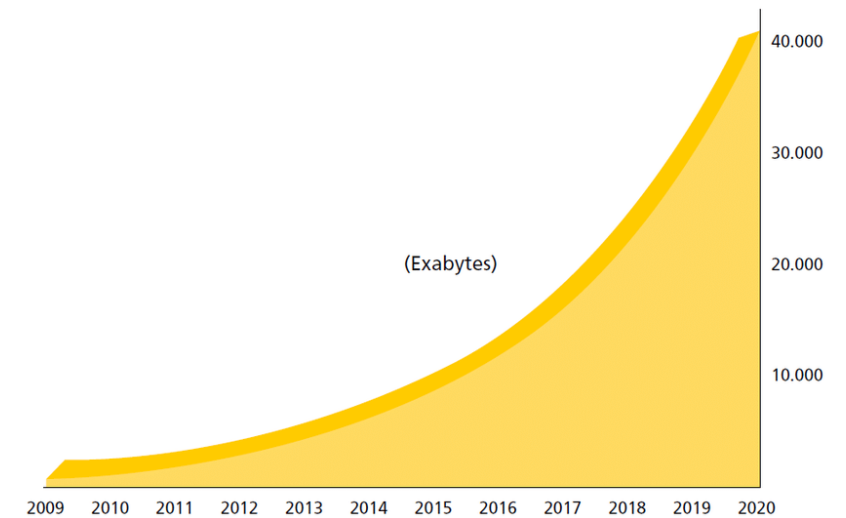
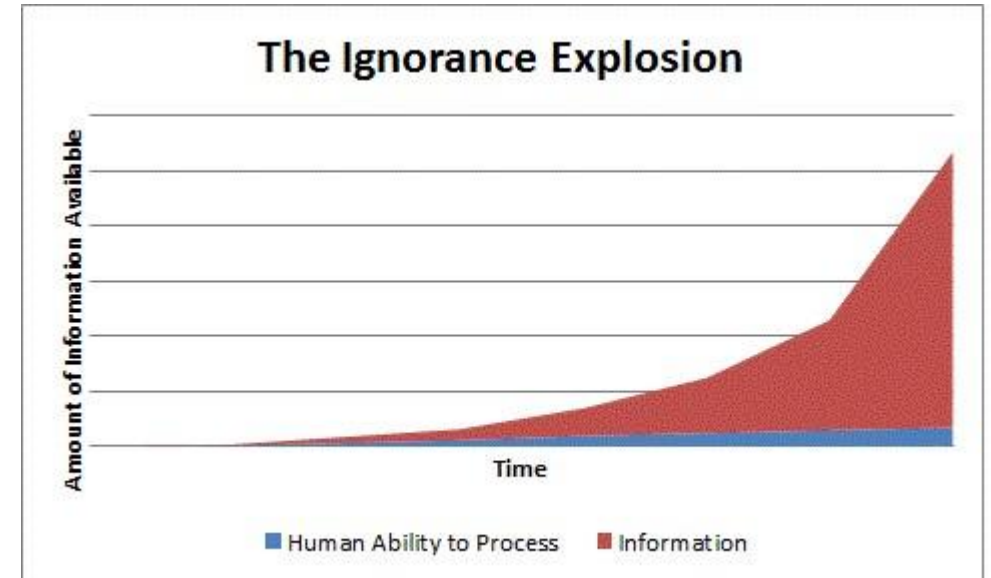
- Each of these issues have their own Big Data challenges.
- Compute, Storage, Speed & Size are all underlying factors in addressing these issues.



The Data Problem

AI Analytics is Required

- The issues surrounding Advanced Cyber Security is not about the analysis it is about the *data*.
- Data management is the cyber security industry's biggest challenge.
- At the heart of this problem that most security solutions do not look at **all the data**.

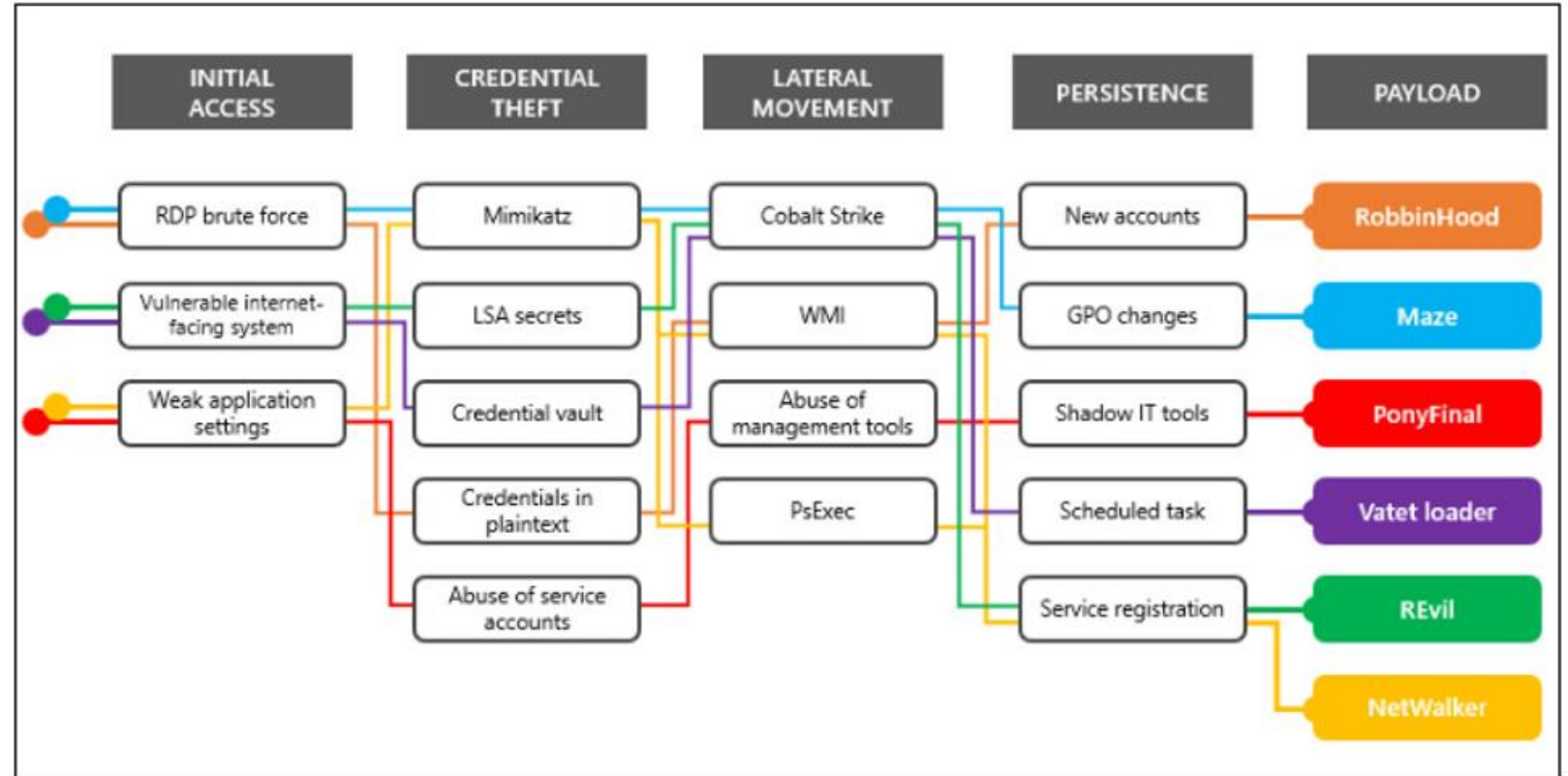


Thread Detection

Thread Detection

Ransomware Attack Stages

1. Initial Access
2. Credential Theft
3. Lateral Movement
4. Persistence
5. Payload (Encryption)



Source: Microsoft

Thread Detection

What to Monitor?

- **Initial Access**

- All Remote Access (firewalls, vpn, authentication, MFA, etc.)
- All Internet Facing Systems
- Application Configuration Changes

- **Credential Theft**

- Login Activity, Privilege Escalation Indicators, Privileged Account Changes

- **Lateral Movement**

- SMB Traffic, Unusual traffic patterns, "first time" behavior, User Behavior

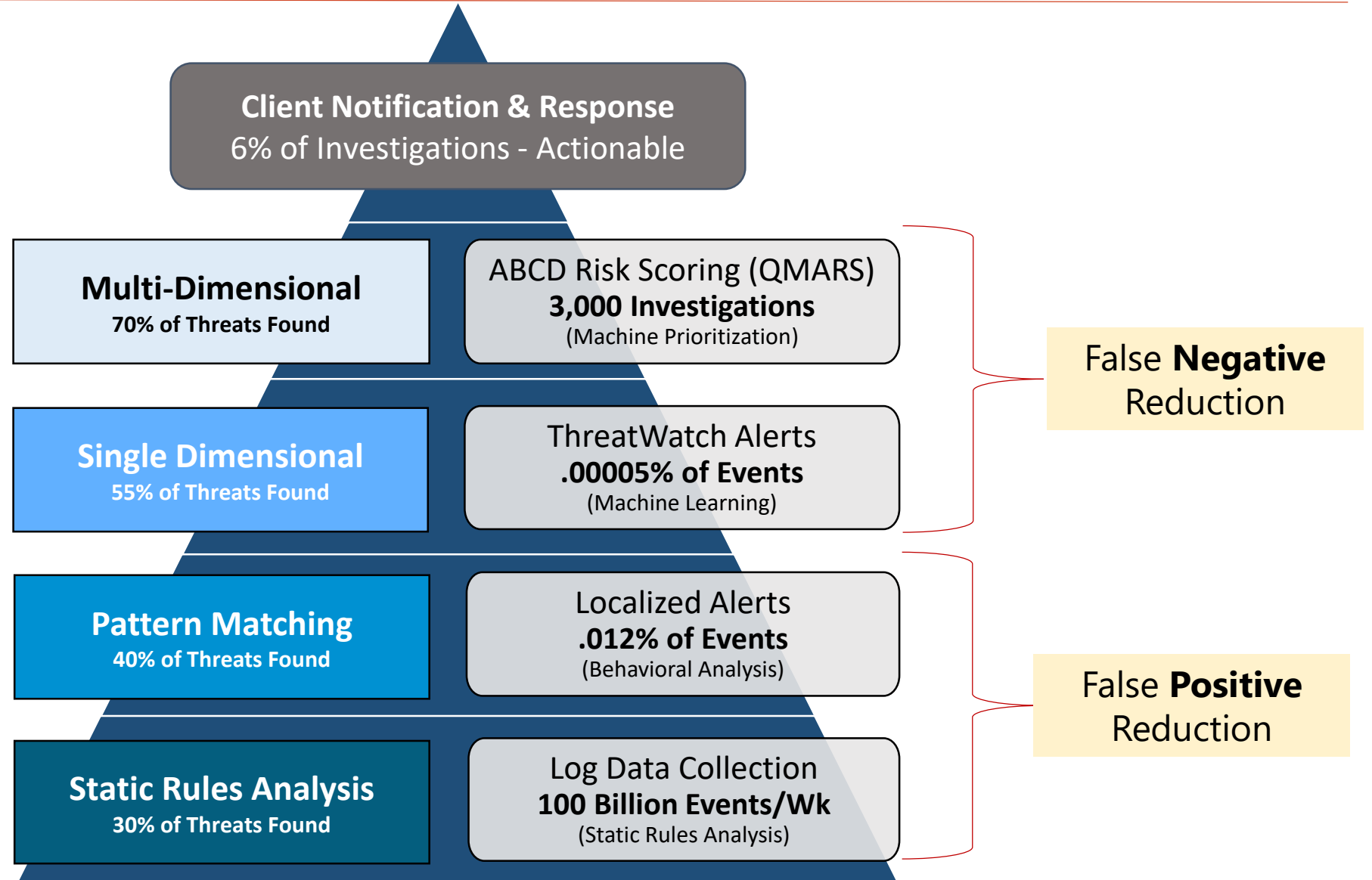
- **Persistence (Back Doors)**

- Account Behaviors/Changes, Beacon Traffic, IP Reputation, Encrypted Comms

- **Payload (Encryption)**

- Any/All Encryption Activity

Thread Detection Analysis Layers

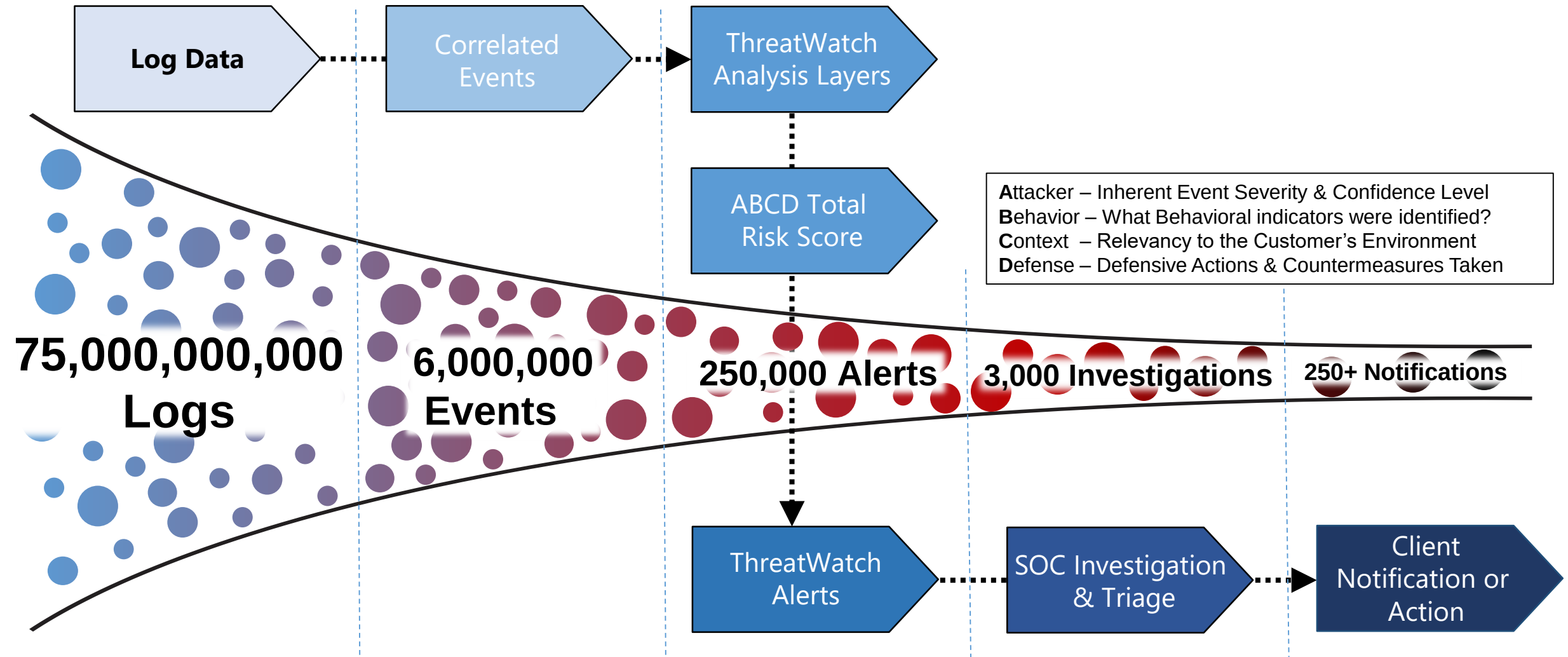


Thread Detection

Analysis Layers

- We can only reduce the amount of data by comprehensive analytics systems.
- This systems often us AI-based algorithms.

Thread Detection Event Processing

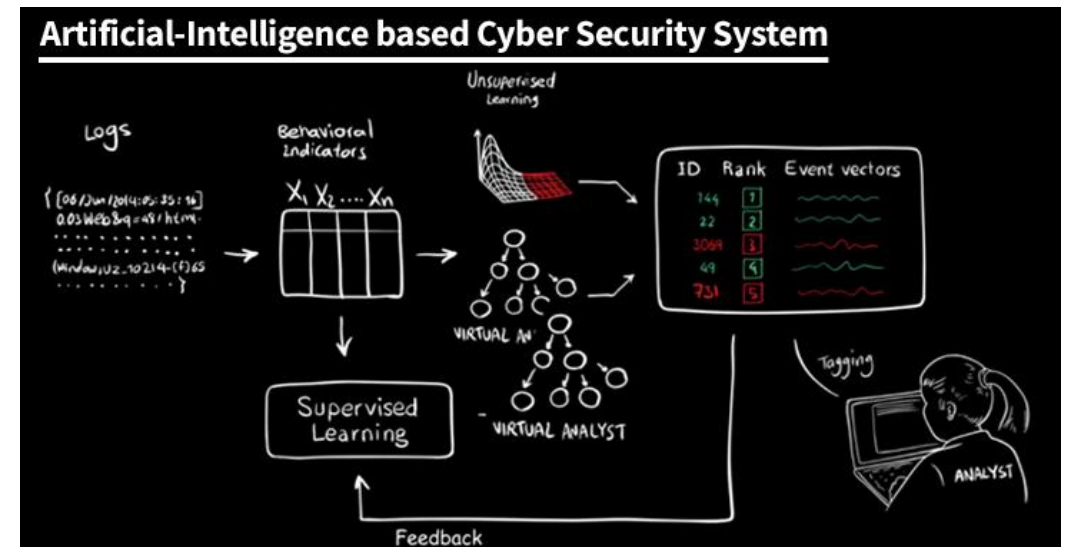


AI in Cyber Security

AI in Cyber Security

Introduction

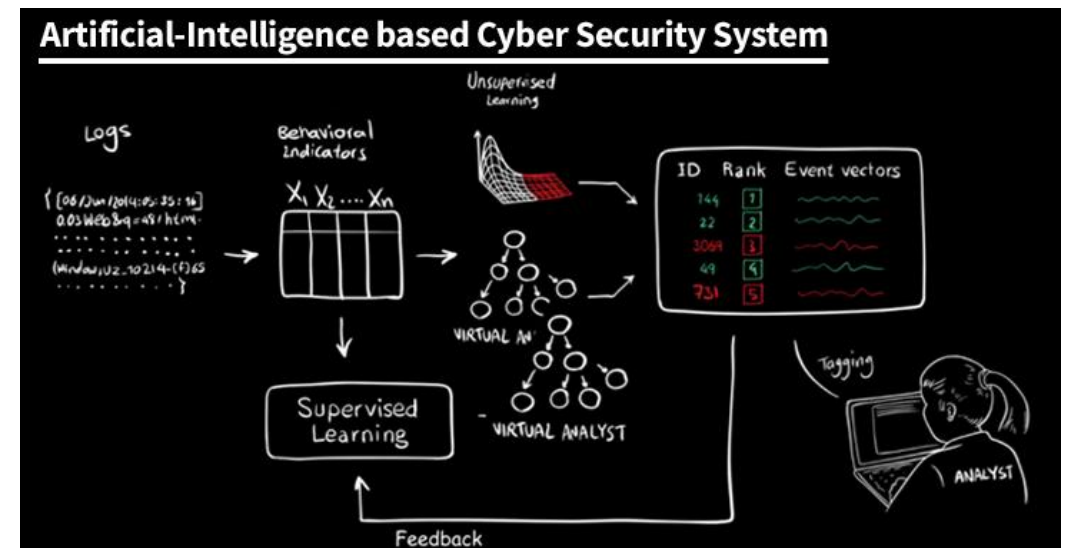
- Almost Everyone in the Cyber-Security Industry is claiming the use of AI & Machine Learning for their product.
- Using AI can improve an organizations ability to respond faster to breaches.
- However, most Threat models which incorporate AI systems are still not mature.
- The security industry is still at the beginning using AI.



AI in Cyber Security

Bad Guys are also Using Machine Learning

- Just because the product vendors are using it, doesn't the bad guys are not.
- The bad guys have a machine that is learning vs your machine that is learning.
- For example, just last year, criminals employed AI-based software to replicate a CEO's voice to command a cash transfer of €220,000 (approximately \$243,000).



AI in Cyber Security

Red and Blue AI

- **AI versus AI cybersecurity** competition [DARPA "Grand Challenge"](#)
 - 2013-2016 – 55M\$ event, 3,75M\$ prize
 - Winner: [Mayhem](#) – its code is now used by the [Pentagone](#) and [US Defense in war zones](#)
 - Attack, Defend (Run services and Patch (which created most of problems))
- Red AI is **AI based Cybercrime**
 - Next Level Threats (NLT's) based Cyberattacks. This means cyberattacks which are using artificial intelligence, machine learning and robotics in combination to make attacks even harder to detect and fight against.
 - Difficult to assess the level of this risk
 - Most example in the next slide below are at the research stage
 - Could some be already available from the Darkweb ?
- Blue Ai is **Defensive AI**

AI in Cyber Security

Red AI

- H – S • **Malware creation:** speed up creation & Enhance evasive capabilities
 - Customized undetectable malware using Elon Musk's OpenAI (2017 Defcon)
 - Extension on polymorphic malware: modify code on the fly based on how and what has been detected in the environment
- M – M • **Smart botnets**
 - Self learning botnets: actions based on local intelligence and exchanges between botnets
 - Smarter zombies: act without the botnet C&C instructions
- H – S • **Advanced spear phishing:** text-to-speech, speech recognition, and natural language processing (NLP) for smarter social engineering
 - Train on genuine emails and make convincing scams
 - “Automated End2End spear phishing on Twitter”: success rate varying between 30 and 60 % (Black Hat USA 2016)
- M – M • **Counter threat intelligence**
 - DDoS TI: raising the noise floor generates a lot of false positives to common machine learning models → once a target recalibrates its system to filter out the false alarms, the attacker can launch a real attack that can get by the defensive ML
- M – S • **Unauthorised access:** Breaking current CAPTCHA (98% success)
- M – S • **Poisoning machine learning engines**
 - 2017: convolutional neural networks (CNNs) attacked to produce false (but controlled) results through CNNs like Google, Microsoft, and AWS
- M – S • Using AI to **classify victims** and optimize RoI
- ? – S • **Condition based Cyberattacks** e.g. Cyberattacks using Blockchain based smart contracts

Estimations

H(igh) – M(edium) – L(ow) risk
S(hort) – M(iddle) – L(ong) term

AI in Cyber Security

Blue AI

- **AI seems to be the best defense against Red AI**
- Is this a new frontier of information warfare?
 - The instantaneous nature of cyberattacks require an **autonomous defense systems** that could act in an early stage. (e.g. Wannacr, NotPetya etc.).
- This will probably generate an **AI arms race**.
- Resources will be key.

AI in Cyber Security

Blue AI

- **Most Blue AI uses a mix of different technologies such as**
- **Classification** identifies which category a new piece of data belongs to.
- **Anomaly detection** trained on a historical data set - then look unusual activities
 - An anomaly might not be a malicious activity
 - Training needs to be refreshed regularly, since employees, networks, and other systems change over time.
- **Cluster analysis** looks at a much larger variety of factors and behaviors than a human can, and update the clusters in real time.
 - still usually takes a human to look at the clusters or anomalies and determine what they mean

AI in Cyber Security

Blue AI

- **Malware detection:** use large collection of software that's already been divided into malware and legitimate applications, and tell whether a previously unseen app is malicious
 - Google: ML to analyze threats against mobile endpoints
 - Microsoft: multi layered (endpoint & cloud), multi ML endpoint defense
- **Anti Spam**
- **Intrusion Detection & Prevention**
 - Characterizes the network traffic, and automatically generates signatures for blocking the attack
 - Deep neural networks can create a model to detect attacks with strong accuracy
- **Vulnerability management:** identifying, prioritizing and helping to remediate existing vulnerabilities
- **User and Entity Behavior Analytics (UEBA):** based on analysis of network traffic, internal and external behaviors, data access and a wide range of other functions and activities.
- **Data classification:** ease compliance with data privacy and data protection regulations
 - Amazon Macie automates tracking the data to identify, classify and protect sensitive pieces of information such as personally identifiable information (PII), personal health information (PHI), regulatory documents, API keys, secret key material and intellectual property
- **Cyber Threat Intelligence (CTI):** behavior categorization for Threat Intelligence
 - Arizona State University uses machine learning to monitor traffic on the dark web to identify data relating to zero-day exploits
- New generation of **honeypots**

AI in Cyber Security

Blue AI – Emerging Usages

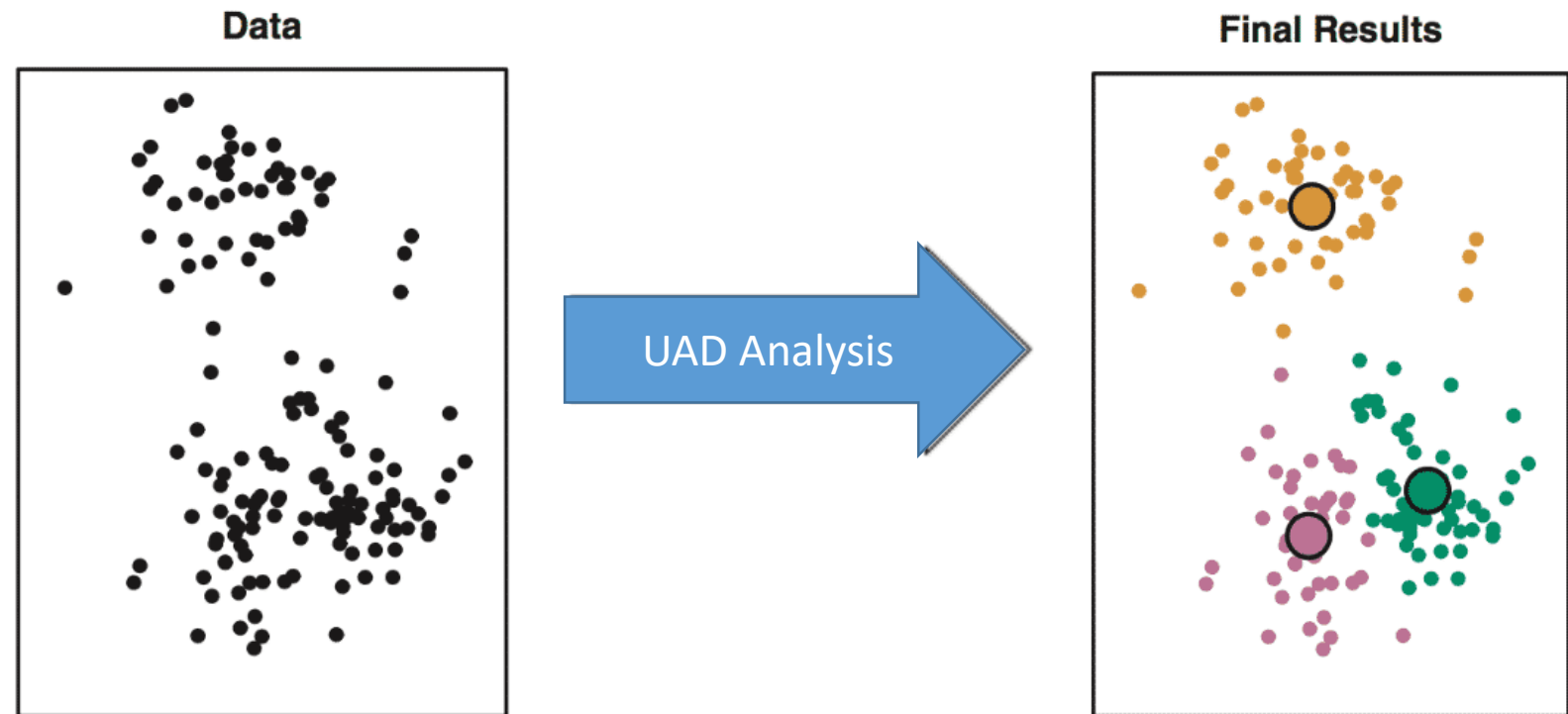
- **Automated remediation**
 - For the recommendations that carry the lowest risk to the company, or that offer the highest benefit
 - DARPA 2016 Grand Challenge
- **Orchestration framework**
 - Allows one security system to trigger an action on a different system, without requiring a human intervention (e.g. log into individual systems and manually execute commands)
 - Will demand that “basics” practices are in place, such as security playbooks
- **Intrusion Detection & Prevention – Autonomous incident response**
 - Watch as security professionals deal with security issues
 - Gather enough historical data to make meaningful predictions
- **Intrusion Detection & Prevention – Automatic self assessment and remediation:** learn attack behaviors that can evade Intrusion Detection Systems (IDS), thereby *enabling the automated creation of data from which an IDS system can learn*
- **Automated pen-testing**
- **Blockchain** could enhance data integrity, digital identities, enable safer **IoT** devices to prevent DDoS attacks, etc.

AI in Cyber Security

Unsupervised ML to Find Unknown Threats

- No Pre-Knowledge Required
- No Baselines
- No Data Bias
- No Human Bias
- No wrong answers
- Training is minimal

Discerning Patterns in the Data without Knowing What to Look For



Finding Patterns in Big Data where Attackers Masquerade as Normal

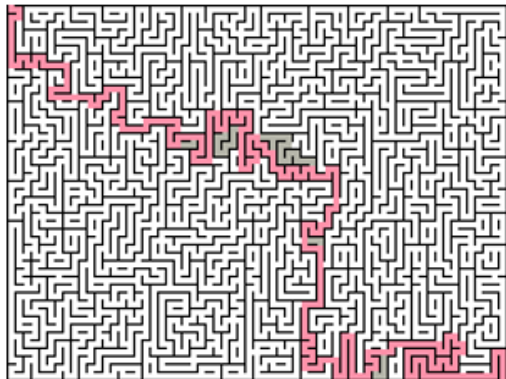
AI in Cyber Security

Unsupervised ML to Find Unknown Threats

- SIEM analyzes data along previously defined pathways. You know how to find the threat (get through the maze) based on pre-programmed logic.
- UAD analyzes all the pathways & optimizes the best path to the threat, without pre-knowledge

The Old Way (SIEM)

Analyze a finite set of paths traveled



The New Way (Threat Analytics)

Analyze infinite destinations and paths



Full Spectrum Analysis (UAD)

Analyze every path & find combinations that are unique or unusual to find the threat



UAD

1. Learns all the pathways
2. Analyzes all combinations of paths together
3. Notes a deviation or pattern change "on the fly"
4. Scores Risk



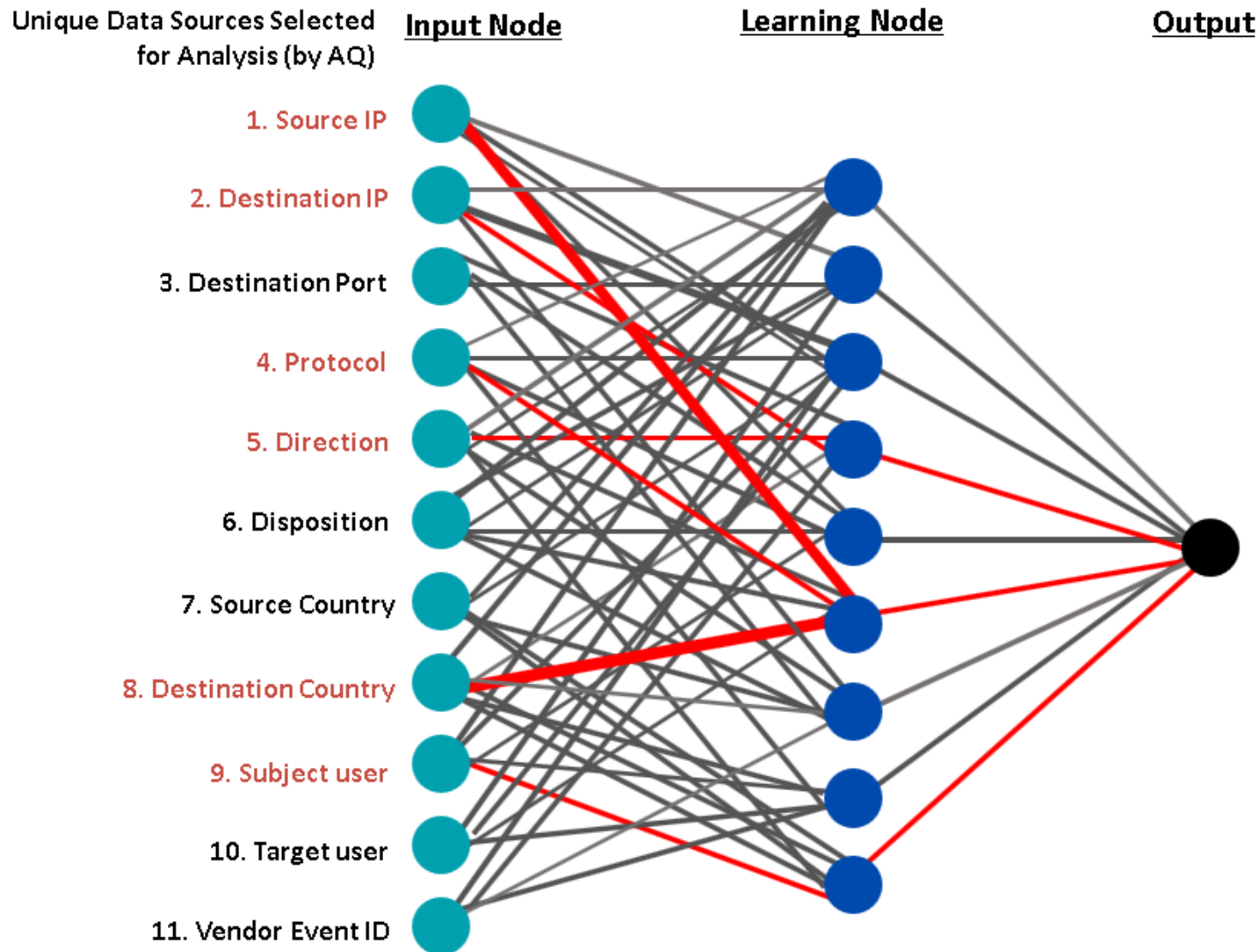
AI in Cyber Security

Variability of Log Data by Data Source

Data Types (11)	Total Possible Values	Total Combinations Possible (From 11 Data Sources) That Require Analysis
Source IP	4,228,250,625	7,412,123,345,625
Destination IP	4,228,250,625	31,340,315,168,716,000,000,000
Destination Port	65,656	2,057,679,732,717,220,000,000,000,000
Protocol	50	102,883,986,635,861,000,000,000,000,000
Direction	10	1,028,839,866,358,610,000,000,000,000,000
Disposition	6	6,173,039,198,151,650,000,000,000,000,000
Source Country	240	1,481,529,407,556,400,000,000,000,000,000
Destination Country	240	355,567,057,813,535,000,000,000,000,000,000
Subject user	1,000	355,567,057,813,535,000,000,000,000,000,000,000
Target user	1,000	355,567,057,813,535,000,000,000,000,000,000,000,000
Vendor Event ID	56,000	19,911,755,237,558,000,000,000,000,000,000,000,000,000,000

AI in Cyber Security

Deep Learning Example



6 Data Column Combination

- Source IP, Destination IP, Destination Country, Protocol, User, Direction

1. Src IP = 232.4.33.1
2. Dst IP = 48.21.128.29
4. Protocol = RDP (3389)
5. Direction = Internal to External
8. Destination = China
9. Subject User = Admin

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Important Questions

- What Model Is your vendor using: static or dynamic?
- How many ML models are in the system?
- If dynamic, how often is the model updated (re-trained)?
- How much data do they use to train the models? (All or subset?)
- How often is that data refreshed (periodic or continuous)?
- How do you deal with unknown conditions or things that were not a part of the training data?